

TAUFIQUL HAQUE KHAN TUSAR

AI/ML RESEARCHER · AI AND BACKEND ENGINEER

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Education

Bachelor of Science (B.Sc.) in Computer Science & Engineering

City University

Apr 2018 – Dec 2022

Dhaka 1216, Bangladesh

- **Thesis:** “Automated Detection of Acute Lymphoblastic Leukemia Subtypes from Microscopic Blood Smear Images Using Deep Neural Networks,” doi: [10.48550/arXiv.2208.08992](https://doi.org/10.48550/arXiv.2208.08992)
- **Research Focus:** Natural Language Processing, Healthcare AI, and Accessible AI.
- **Research Achievements:** 4 peer-reviewed publications (1 journal, 3 conferences).
- **CGPA:** 3.21 / 4.00

Publications

Journal: 1 • **Conference:** 3 • **Preprint:** 1 • **Citations:** 185+ as of Aug 2026 • **h-index:** 5

- [1] **Taufiqul Haque Khan Tusar** et al., “An Intelligent Telediagnosis of Acute Lymphoblastic Leukemia using Histopathological Deep Learning,” Journal of Computing Theories and Applications (JCTA), vol. 2(1), pp. 1–12, 2024. doi: [10.62411/jcta.10358](https://doi.org/10.62411/jcta.10358).
- [2] J. Hossain, M. T. Islam, and **Taufiqul Haque Khan Tusar** “Streamlining Brain Tumor Classification with Custom Transfer Learning in MRI Images,” 2023 IEEE SIST, Astana, Kazakhstan, 2023, pp. 522-526, doi: [10.1109/SIST58284.2023.10223507](https://doi.org/10.1109/SIST58284.2023.10223507).
- [3] **Taufiqul Haque Khan Tusar**, M. T. Islam, and A. H. Sakil, “An experimental study for early diagnosing Parkinson’s disease using machine learning,” arXiv preprint, 2023. doi: [10.48550/arXiv.2310.13654](https://doi.org/10.48550/arXiv.2310.13654).
- [4] **Taufiqul Haque Khan Tusar**, M. T. Islam and F. I. Raju, “Detecting Chronic Kidney Disease (CKD) at the Initial Stage: A Novel Hybrid Feature-selection Method and Robust Data Preparation Pipeline for Different ML Techniques,” 2022 5th IEEE ICCI, Cairo, Egypt, 2022, pp. 400-407, doi: [10.1109/ICCI54321.2022.9756094](https://doi.org/10.1109/ICCI54321.2022.9756094).
- [5] **Taufiqul Haque Khan Tusar** and M. T. Islam, “A Comparative Study of Sentiment Analysis Using NLP and Different ML Techniques on US Airline Twitter Data,” 2021 IEEE ICECIT, Khulna, Bangladesh, 2021, pp. 1-4, doi: [10.1109/ICECIT54077.2021.9641336](https://doi.org/10.1109/ICECIT54077.2021.9641336).

Research Experience

Research Engineer

AIMS Lab, United International University

Feb 2023 - Mar 2023

Dhaka 1212, Bangladesh

- Collaborated with the research team to transform research outputs into integrable applications.
- Developed the Flask backend for a multimodal AI receptionist robot.

Undergraduate Research Assistant

Supervisor: Md. Touhidul Islam, Lecturer, Dept. of CSE, City University

Dhaka 1216, Bangladesh

Dec 2020 - Nov 2022

- Worked on machine learning research in NLP, statistical machine learning, and medical image processing.
- Co-authored 4 peer-reviewed publications and presented 2 accepted papers at academic conferences.
- Contributed to paper writing, literature review, data preparation, and experiments.
- Mentored junior students in machine learning concepts and research work.
- Deployed two machine learning models as Flask web applications.

Industry Experience

AI and Application Engineer

KafKai Giken Inc.

May 2023 - present

Tokyo 110-0016, Japan (Remote)

- Built and maintained the Django-PostgreSQL service and AI agents for [KAFKAI V3](#).
- Developed Ninja API endpoints and handled long-running tasks using Redis and Celery.
- Reduced LLM API costs by routing tasks to appropriate models and providers via OpenRouter.
- Worked on the FastMCP server, implemented action-based credit usage, and integrated Stripe top-ups.

- Fine-tuned and maintained open-weight LLMs using Transformers, PEFT, and PyTorch for domain adaptation.
- Maintained the legacy Pyramid-based codebase and migrated it to a Django-PostgreSQL architecture.
- Managed production infrastructure on AWS Lightsail, PostgreSQL databases, and deployments.
- Collaborated to improve UI/UX of [Kafkai.com](https://kafkai.com), [Kafkai.ai](https://kafkai.ai), and [Puchiden.app](https://puchiden.app).
- Managed affiliate program communications and outreach for [Puchiden.app](https://puchiden.app).
- Conducted technical interviews and supported the hiring process for the engineering team.

AI Product Engineer

Feb 2026 - Present

ChatCopilot.app

Gazipur, BD · Independent Venture

- Architected and built ChatCopilot.app, a multi-tenant Django SaaS for agentic AI-based sales, support, and appointment.
- Implemented the ReAct agent and agent orchestration, including tool calling, state management, compaction, routing, translation, image understanding, audio transcription, and human handover.
- Built tenant-scoped RAG for business knowledge and product/service catalogs using pgvector and multimodal embeddings.
- Designed integrations for Meta Graph API, Google Calendar, custom ERP calendars, webhooks, and Pathao courier dispatch.
- Managed the DigitalOcean production droplet, PostgreSQL database, and deployment.

Machine Learning Engineer (Intern)

July 2022 - Aug 2022

Shadhin Lab LLC

Dhaka 1219, Bangladesh

- Worked on NLP-based, large-volume data collection, labeling, and classification pipelines for an in-house product.

Teaching Experience

Teaching Assistant

Gazipur 1750, Bangladesh

Milton English & Math Academy

Nov 2019 - Mar 2020

- Taught Physics and ICT to secondary and higher-secondary students.

Academic & Technical Services

- **Reviewer**, EuroPython 2025, Prague, Czechia (14-20 July 2025)
- **Reviewer**, 3rd International Conference on Computing and Information Technology (ICCIT), 2023, Tabuk, Saudi Arabia

Conferences

- **Speaker**, PyCon ID 2025, Jakarta, Indonesia (14-15 December 2025)
- **Presenter**, 5th International Conference on Computing and Informatics (ICCI), 2022, Cairo, Egypt
- **Presenter**, International Conference on Electronics, Communications and Information Technology, 2021, Khulna, Bangladesh

Achievements

- **Best Paper Award**, 2023 IEEE International Conference on Smart Information System and Technologies
- **Merit Scholarship**, City University, Bangladesh — Summer 2020
- **Junior School Government Scholarship**, Ministry of Education, Bangladesh — 2012

Technical Skills

Languages	Python, SQL, Bash, C
AI Engineering	Transformers, PyTorch, FastMCP, llama-server, llama-cpp-python, pgvector
LLM APIs/Tools	OpenAI, Anthropic, OpenRouter, Perplexity, JinaAI, Ollama
ML & Data	scikit-learn, NLTK, OpenCV, pandas, NumPy, SciPy, Matplotlib
Backend	Django, Flask, Celery, Redis, Pytest, PostgreSQL, SQLite, Stripe
Deployment	AWS (EC2, Lightsail, S3), DigitalOcean, Render, Cloudflare (R2, Workers, D1), Caddy
Others	Git & GitHub, Linux, LaTeX, HTML

Open Source

• MochaChat

Multi-character AI companion with streaming conversations. Built a stateless FastAPI backend with local per-character browser memory, LLM memory compaction, multilingual support, and cost-efficient LLM routing via OpenRouter.

Tools: FastAPI, OpenRouter, Render

- **Python Buddy**

Fine-tuned a small language model (Qwen2.5-Coder-1.5B-Instruct) for code assistance and worked on low-resource training and inference. Achieved 7–12 tokens/s inference speed with 8-bit quantization on Snapdragon 685.

Tools: Hugging Face Transformers, llama.cpp, Python

- **Inference API Server**

Developed a lightweight Python API server for local/self-hosted LLM inference. Supports GGUF models with both string-based and role-based prompts, and returns OpenAI-compatible responses.

Tools: Python, Hugging Face Hub, llama-cpp-python

- **Blood Cancer Stage Detection App**

Built a deep learning web app for acute lymphoblastic leukemia (ALL) stage detection. Fine-tuned and evaluated 4 CNN architectures; MobileNetV2 achieved 98% accuracy across 3 leukemia subtypes.

Tools: TensorFlow CPU, Flask, OpenCV, Unicorn, Render

- **Chronic Kidney Disease Diagnosis App**

Built a Flask-based ML app for early chronic kidney disease detection. Used LOF for outlier removal and SMOTE for imbalance handling; Random Forest achieved the highest accuracy on the test data.

Tools: Scikit-learn, Flask, Unicorn, Joblib, Render

Community & Outreach

Community Lead

AI/ML Professional Community Bangladesh

March 2025 - Present

Bangladesh

- Organize technical webinars and workshops on AI/ML applications.
- Mentor early-career practitioners in development and career pathways.

Community Organizer

Python Bangladesh

Jan 2025 - Dec 2025

Bangladesh

- Organized tech meetups and collaborative events for Python developers across Bangladesh.
- Led initiatives to restart PyCon Bangladesh.

Data Science Bootcamp Mentor

AI Planet (Formerly DPHI)

Sept 2021 - Oct 2021

Hyderabad, India (Remote)

- Supported the core team in running the data science bootcamp.
- Guided participants on exploratory data analysis and created and evaluated final assessments.

Memberships

- **Member**, IEEE Computer Society (Aug 2022 - Dec 2022)
- **Student Member**, Institute of Electrical and Electronics Engineers (IEEE) (Aug 2021 - Dec 2022)
- **Executive Member**, City University Computer Club (Dec 2020 - Oct 2022)

Extracurricular Activities

- **Student Ambassador**, IEEEExtreme (Aug 2022 - Oct 2022)
- **Chapter Influencer**, Google Crowdsource (Sep 2021 - Jun 2022)
- **V-Force Volunteer**, United Nations Volunteer (Aug 2021 - Sep 2021)

References

- **Kamal Mustafa** —  kamal@kafkai.com
CTO, Kafkai Giken Inc., Tokyo 110-0016, Japan
- **Md Touhidul Islam** —  touhid000cse@gmail.com
Sr. Software Engineer, Samsung R&D Institute, Dhaka, Bangladesh
Former Lecturer, Dept. of CSE, City University, Dhaka 1216, Bangladesh
- **Brig. General (Rtd.) Prof. Dr. Engr. Md. Lutfor Rahman** —  lutfor.mist@yahoo.com
Dean, Faculty of Science & Engineering, City University, Dhaka 1216, Bangladesh